

High Performance Climate and Weather Benchmark (HPCW)

Correctness and Reproducibility for Earth System Software

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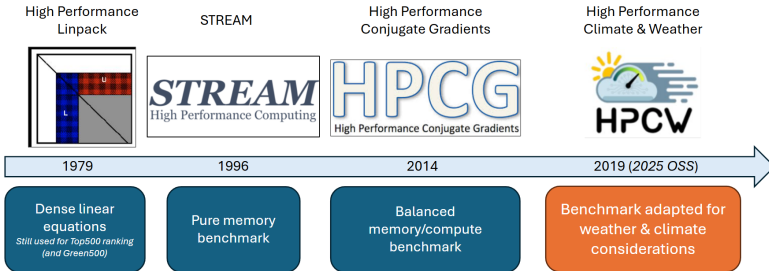


Agenda

- 1 What is HPCW?
- 2 Reproducibility
- 3 Usage Examples
- 4 Summary

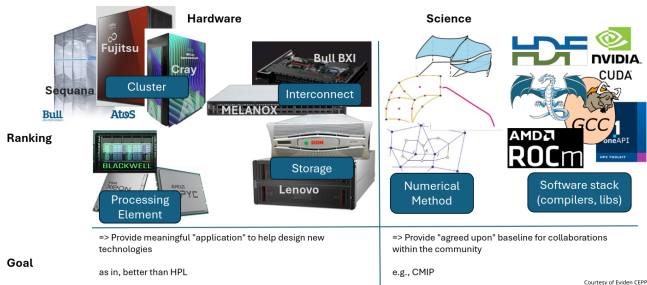
Making the Case for Another Benchmark

HPC Benchmarks - Overview

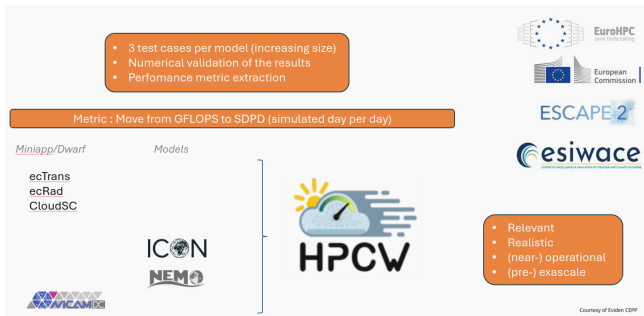


Courtesy of Eiden CEPP

Making the Case for Another Benchmark



High-level Overview



Components of a Benchmark

- 1 Code to benchmark
- 2 Test case
- 3 Validation
- 4 Metrics

Components of a Benchmark

- 1 Code to benchmark
 - Provide everything in a CMake "Superbuild"
 - Pinned versions of the benchmark applications
 - Pinned versions of the dependencies (optional)
- 2 Test case
- 3 Validation
- 4 Metrics

Components of a Benchmark

- 1 Code to benchmark ✓
- 2 Test case
 - All arguments and input files are provided
- 3 Validation
- 4 Metrics

Components of a Benchmark

- 1 Code to benchmark ✓
- 2 Test case ✓
- 3 Validation
 - Bitwise reproducibility obviously not an option
 - Instead: numerical validation, specialized per application/test case
 - Comparison to reference values $\pm$ tolerance, generated via expert knowledge, known solutions or ensembles
- 4 Metrics

Components of a Benchmark

- 1 Code to benchmark ✓
- 2 Test case ✓
- 3 Validation ✓
- 4 Metrics
 - Run time as bare minimum
 - Experimental support for CPMIP metrics (CHSY, Complexity etc.)

Example 1: EPI

- Project goal: European independence in High Performance Computing Processor Technologies
- Needs applications to verify a system's usefulness
- Currently using HPCW to evaluate
 - the capabilities of ARM systems to compete with "traditional" x86
 - the LLVM-based compilers that are used in the context of this project

Example 2: System Testing / Performance Monitoring

- DKRZ provides software to its users, requires regular updates
 - Do not want regressions
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- ⇒ Automatic monitoring with HPCW in a matter of days
 - ⇒ Reproducibility guarantees from HPCW
 - ⇒ Covers many different software packages due to amount of dependencies

Summary

- HPCW as domain-specific benchmark suite
- Focus on reproducibility (in a benchmarking context)
- Fully open-source, go to the [HPCW repository](#)

Acknowledgment



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